Payzo Books: Enterprise Cloud Accounting & ERP Platform

Master Technical Handbook: Beginner to Extreme Senior Architecture

Project: Payzo Books Enterprise ERP
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Tech Stack: Flutter, Dart, Riverpod, SQLite/Drift, ZATCA Phase 2 E-Invoicing, Android MediaStore Bridge, Excel/PDF Generation, 792 Dart Files

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Section 1: End-to-End ERP Architecture & Working Explanation

1.1 Beginner Level: What is Enterprise Accounting Software?

Small and medium businesses must track every financial movement: issuing sales invoices to clients, recording vendor bills, tracking payments, calculating taxes (VAT / GST), and managing inventory. If an accounting software is slow, loses an invoice, or calculates 1 cent incorrectly, the company faces tax penalties or audits.
Payzo Books provides complete business accounting directly on mobile and web with 792 Dart files structured into Clean Architecture modules.

1.2 Extreme Senior Level: Modular Codebase & Riverpod State Architecture

With 792 files, monolithic state management creates unmaintainable spaghetti code. Payzo Books implements **Clean Architecture with Feature-First Modularization**:

```
lib/
├── core/                  # Database, Network Client, Storage, Theme, Currency
│   ├── database/          # Drift (SQLite) ORM with Type Converters
│   ├── network/           # Dio Interceptors, SSL Pinning, Retry Logic
│   └── errors/            # Failure models & Result<T> monads
└── features/
    ├── invoices/          # Invoices, Quotations, Credit Notes
    ├── bills/             # Vendor Bills, Expenses, Purchase Orders
    ├── inventory/         # Stock Items, Low Stock Alerts, Warehouses
    ├── zatca/             # Saudi ZATCA Phase 2 XML/QR Generator
    └── reports/           # Profit & Loss, Balance Sheet, VAT Return
```

Riverpod Dependency Graph

- Uses `AutoDisposeNotifierProvider` for form state so invoice line-item memory is reclaimed upon closing the editor.
- Uses `FutureProvider.family` for cached ledger lookups with background revalidation.
- `ref.watch(invoiceProvider.select((inv) => inv.totalAmount))` ensures that adding a line item updates only the summary bar without re-rendering individual table inputs.

1.3 Financial Arithmetic: The Floating-Point Catastrophe

Standard IEEE 754 floating-point math (`double` in Dart) cannot represent decimal fractions like `0.1` precisely in binary:

```
// The Double Trap:
double price = 100.10;
double vatRate = 0.15;
print(price * vatRate); // Prints 15.015000000000002 -> Fails tax audit!
```

Payzo Books Solution: All monetary amounts are stored as integer **Minor Currency Units** (cents / fils / halalas) or processed via the `decimal` package:

```
final subtotal = Decimal.parse("100.10");
final vat = (subtotal * Decimal.parse("0.15")).toDecimal(scaleOnInfinitePrecision: 2);
```

1.4 ZATCA Phase 2 E-Invoicing Compliance (Saudi Tax Authority)

For Saudi Arabia compliance, every B2B and B2C invoice requires: 1. **Cryptographic Stamp:** An ECDSA secp256k1 digital signature over the invoice UBL 2.1 XML content. 2. **Invoice Hash:** SHA-256 hash of the normalized XML document chained to the previous invoice's hash (forming a tamper-evident blockchain-style ledger). 3. **ZATCA QR Code:** A Base64-encoded TLV (Tag-Length-Value) data structure containing Seller Name, VAT Number, Timestamp, Total with VAT, and Cryptographic Stamp.

1.5 Android MediaStore & Scoped Storage Bridge

In modern Android (API 29+ Android 10, 11, 12, 13, 14), accessing raw file paths via `dart:io` in public storage throws `PermissionDeniedException`.
How Payzo Books Exports Invoices & Excel Reports: - Implemented a custom Platform Channel invoking Android's `MediaStore` **API**: 1. PDF/Excel is generated in the app's sandboxed cache directory. 2. Kotlin platform channel creates a new entry in `MediaStore.Downloads.EXTERNAL_CONTENT_URI` with `IS_PENDING = 1`. 3. Writes bytes into the opened `ContentResolver.openOutputStream()`. 4. Flushes and marks `IS_PENDING = 0`, making the invoice immediately visible in the device's Downloads folder without requiring dangerous storage permissions.

Section 2: Code Audit Findings, ERP Architecture Risks & Fixes

Finding 1: Memory Leaks in Multi-Page PDF Generation

- **Problem:** Generating a 100-page transaction report with embedded company logos allocated 400MB of RAM, causing low-memory crashes on budget Android devices.
- **Production Fix:** Implemented chunked PDF rendering using an Isolate worker: pages are streamed and compressed to disk incrementally rather than building the complete document in heap memory.

Finding 2: Offline SQLite Sync Conflicts

- **Problem:** A user edited an invoice offline while an accountant edited the same invoice on the web portal. Upon reconnection, offline changes overwrote the cloud.

- **Production Fix:** Added an optimistic concurrency control column `version_id`. If `client.version_id != server.version_id`, the sync engine raises a merge conflict prompt displaying line-by-line diffs.

Section 3: Enterprise Accounting & ERP Q&A

Q1: How do you structure an offline-first sync engine in Flutter?

Answer:
In Payzo Books, all read/write operations execute against local SQLite (Drift). A sync table records every mutation: `entity_id`, `table_name`, `operation` (INSERT, UPDATE, DELETE), `payload_json`, and `sync_status`. A background sync service monitors connectivity. When online, it pushes mutations sequentially inside a single database transaction. If the server accepts, the mutation is cleared; if it fails, it applies exponential backoff retry.

(Handbook continues with full 40 questions covering ZATCA, MediaStore, and large-scale ERP architecture)